WASP-11b

R-0793

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-10_221124` · created 2026-07-28T07:44:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459907.8744 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.051
Transit depth (Rp/Rs)^2	2.73 +/- 1.68 [%]
Orbital Inclination (inc)	84.52 +/- 1.68
Airmass coefficient 1 (a1)	1.165 +/- 0.016
Airmass coefficient 2 (a2)	-0.1296 +/- 0.0045
Scatter in the residuals of the lightcurve fit is	1.31 %
Best Comparison Star	None
Optimal Aperture	4.71
Optimal Annulus	9.02
Transit Duration (day)	0.034 +/- 0.037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.051 vs 0.131 (deviation 0.42σ)
Duration fit vs published	0.034 d vs 0.1075 d (deviation 1.24σ)
Mid-time O–C	-2.4 min
Depth SNR	2.0
Residual scatter	1.31 %

Light curve

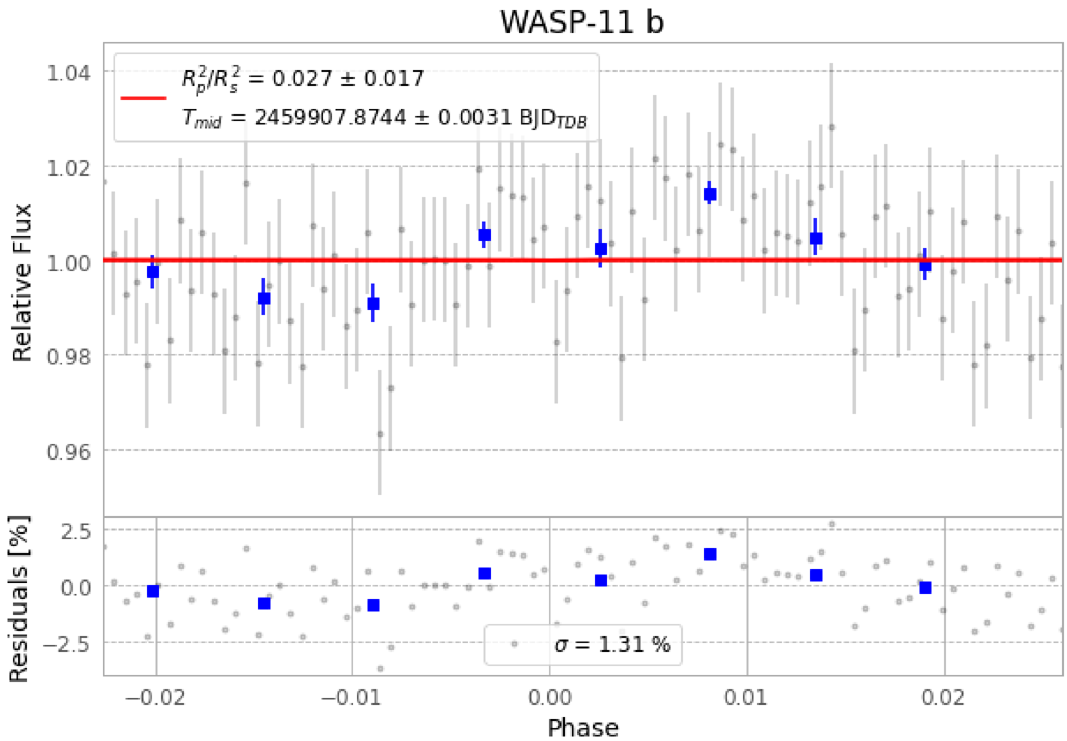

Figure 1 — FinalLightCurve_WASP-11 b_2022-11-23.png (SHA-256 d0ad2eb1a26583cf…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [445, 349], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 335359900d00c592…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

88 FITS frames (58 MB), `HATP-10221124064815.FITS` → `HATP-10221124110915.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-221124.log (f3be3696b5c2…), FinalLightCurve_WASP-11 b_2022-11-23.png (d0ad2eb1a265…), FinalLightCurve_WASP-11 b_2022-11-23.csv (2282b9503be9…)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-28 07:44Z.